



Modelling with Differential Equations

• Worksheet 1

Core Pure 2 Chapter 8: Modelling with Differential Equations. Answer all questions. Show your working clearly. Total: 70 marks. Time: 90 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 8. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Time t is measured in the units stated in each question. Give answers to 3 significant figures unless an exact answer is asked for.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Forming differential equations from rate statements	22	
B First-order models	21	
C Second-order models: SHM, damping and interpretation	22	
Roar-level challenge	5	
Total	70	

Section A Forming differential equations from rate statements

22 marks

1. The rate of increase of a population P at time t is proportional to the size of the population. [2]
Write down a differential equation connecting P and t , stating the sign of any constant you use.

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2. A rumour spreads through a school of 1000 students. The rate at which N , the number of students who have heard the rumour, increases is proportional to the product of the number of students who have heard it and the number who have not. [3]

- (a) Write down a differential equation for N in terms of t .
(b) Find the value of N at which the rumour spreads fastest.

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3. A tank holds 100 litres of salt solution. Brine containing 0.2 kg of salt per litre flows into the tank at 4 litres per minute. The solution is kept well mixed and flows out at 4 litres per minute. [3]
Given that the tank contains x kg of salt at time t minutes, show that $\frac{dx}{dt} = 0.8 - \frac{x}{25}$.

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4. A spherical snowball melts so that its volume decreases at a rate proportional to its surface area. [4]
(a) Show that the radius r cm of the snowball decreases at a constant rate.
(b) The radius falls from 6 cm to 4 cm in 2 hours. Find how much longer the model predicts it will take for the snowball to melt completely.

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5. A tank initially contains 50 litres of pure water. Brine containing 0.5 kg of salt per litre flows into the tank at 3 litres per minute, and the well-mixed solution flows out at 2 litres per minute. At time t minutes, the tank contains x kg of salt.

(a) Show that

[3]

$$\frac{dx}{dt} = 1.5 - \frac{2x}{50+t}.$$

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(b) Solve the differential equation to find x in terms of t .

[5]

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(c) The tank has a capacity of 150 litres. Find the mass of salt in the tank at the moment it becomes full.

[2]

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(Total for Question 5 is 10 marks)



Section B First-order models

21 marks

6. A population of bacteria satisfies $\frac{dP}{dt} = 0.03P$, where t is in hours, and $P = 2000$ when $t = 0$. [2]
Find the population when $t = 10$.

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7. A cup of tea at 90°C is placed in a room at a constant 20°C . Its temperature $\theta^\circ\text{C}$ at time t minutes is [3]
modelled by Newton's law of cooling,

$$\frac{d\theta}{dt} = -k(\theta - 20).$$

Show that $\theta = 20 + 70e^{-kt}$.

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8. For the model in Question 7, the tea cools to 60°C after 5 minutes. [3]
Find the value of k , and the time taken for the tea to cool to 40°C .

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9. The number of fish, P thousand, in a lake t years after fishing begins is modelled by

[4]

$$\frac{dP}{dt} = 0.2P - 40,$$

with $P = 150$ when $t = 0$. Find P in terms of t , and the time at which the model predicts that the fish will die out.

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10. The population P of a species of bird on an island, t years after the birds were introduced, is modelled by the differential equation

$$\frac{dP}{dt} = \frac{P(1000 - P)}{2000}.$$

Initially there are 100 birds.

- (a) Using partial fractions, show that $P = \frac{1000}{1 + 9e^{-\frac{1}{2}t}}$. [5]

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- (b) Find the time taken for the population to reach 500. [2]

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- (c) State the long-term population predicted by the model, and give one limitation of the model. [2]

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(Total for Question 10 is 9 marks)

**Section C Second-order models: SHM, damping and interpretation**

22 marks

11. A particle moves along a straight line so that its displacement x m from a fixed point O at time t seconds satisfies $\frac{d^2x}{dt^2} = -9x$. It is released from rest at $x = 4$. Find x in terms of t . [2]

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12. A particle moves with simple harmonic motion about O , with amplitude 0.5 m and period 4 seconds. Find the maximum speed and the maximum magnitude of acceleration of the particle. [3]

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13. The motion of a particle attached to a spring and moving through a liquid is modelled by [3]

$$\frac{d^2x}{dt^2} + k\frac{dx}{dt} + 16x = 0,$$

where k is a positive constant. State, with a reason, the type of damping when $k = 10$, when $k = 8$ and when $k = 2$.

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14. A damped oscillation is modelled by

[4]

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0,$$

with $x = 2$ and $\frac{dx}{dt} = 0$ when $t = 0$. Find x in terms of t .

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15. The numbers of rabbits, x , and foxes, y , in a region at time t years are modelled by

$$\frac{dx}{dt} = 0.2x - 0.2y, \quad \frac{dy}{dt} = 0.1x.$$

Initially there are 100 rabbits and 20 foxes.

(a) Show that $\frac{d^2x}{dt^2} - 0.2\frac{dx}{dt} + 0.02x = 0$. [3]

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(b) Find the general solution for x . [2]

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(c) Find x in terms of t . [3]

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(d) Find the time at which the model predicts that the rabbits die out. [2]

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(Total for Question 15 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A particle on a spring is driven by a periodic force, and its displacement x m at time t s satisfies [5]

$$\frac{d^2x}{dt^2} + 4x = 3 \sin 2t,$$

with $x = 0$ and $\frac{dx}{dt} = 0$ when $t = 0$.

Find x in terms of t , and describe what happens to the oscillations as t increases. Comment on the validity of the model.

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END OF WORKSHEET

Total: 70 marks